

Candidate Surname Dhrip	Other names Tejas
Date: 23/06/26	
(Time: 60 minutes)	Year Group: 9
Mathematics Year 9 Formals Not tiered	
You must have: Ruler graduated in centimetres and millimetres, protractor, pair of compasses, pen, HB pencil, eraser, calculator Tracing paper may be used.	Total Marks 60

Instructions

- Answer **all** questions.
- You must **show all your working**.
- Diagrams are **NOT** accurately drawn, unless otherwise indicated.
- **Calculators may be used.**



Information

- The total mark for this paper is 75.
- Please try all three sections.

Tick the name of your class teacher

- ☐ Mrs Jackson (9c1/9g1)
- ☐ Miss Chelaghma (9c2)
- ☐ Mr Bartram (9c3)
- ☐ Ms Ye (9c4)
- ☒ Ms Ye and Miss Hort (9g2)
- ☐ Mrs Taylor (9g3)

SECTION 1 MARK	24 / 25
SECTION 2 MARK	20 / 25
SECTION 3 MARK	16 / 25

Turn over ►

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

SECTION 1:

1. a) Simplify: $8w - 4y + w - 3y$

~~8w~~ $9w - 7y$

$9w - 7y$ ✓

[2 marks]

b) Expand: $3(5 - x)$

$15 - 3x$ ✓

[2 marks]

c) Factorise: $3y - 21$

$3(y - 7)$ ✓

[1 mark]

d) Expand $(y + 10)(y + 2)$.

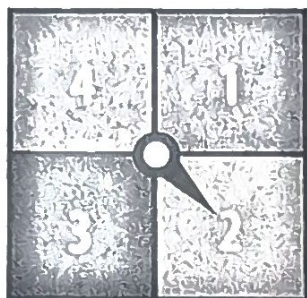
Use the grid to help if you want to.

$\times$	y	10
y	y^2	$10y$
2	$2y$	20

$y^2 + 12y + 20$ ✓

[3 marks]

2. A fair four-sided spinner is spun twice.
The two results are multiplied together as shown in the diagram.
Work out the probability that the score is greater than 7.



×	1	2	3	4
1	1	2	3	4
2	2	4	6	8
3	3	6	9	12
4	4	8	12	16

$$\frac{6}{24} \rightarrow \frac{1}{4}$$

[2 marks]

3. Elysia spun a spinner with four coloured sections 25 times.
The table shows how many times the spinner landed on each colour.

Colour	Purple	Yellow	Green	Red
Frequency	9	1	8	7

- a) Estimate the probability of the spinner landing on green.

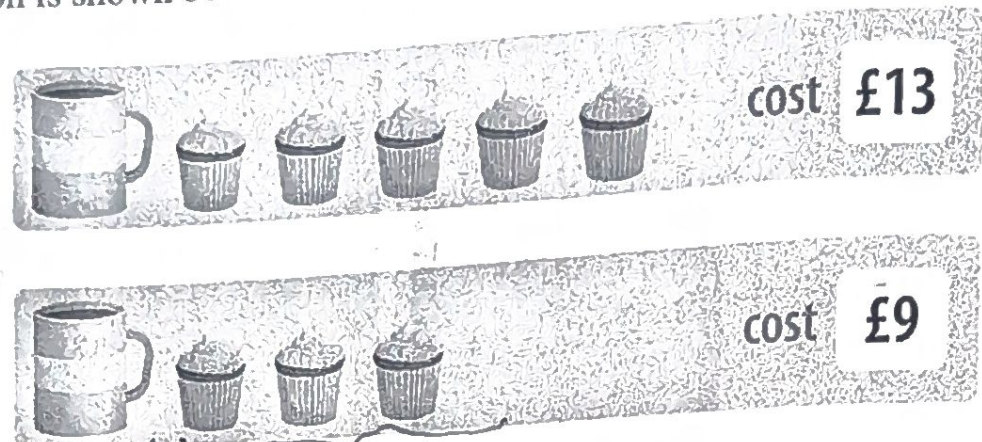
$$\frac{8}{25}$$

[1 mark]

- b) Explain how Elysia could get a better estimate for the probability.

~~Do it~~ Spin the spinner more times
~~maybe 100 times~~ If she spins it a 100 times it would give a much better estimate [1 mark]

4. The cost of 1 cup of tea and 5 cakes is £13
 The cost of 1 cup of tea and 3 cakes is £9
 This information is shown below:



- a) How much do 2 cakes cost? £6

£4

[1 mark]

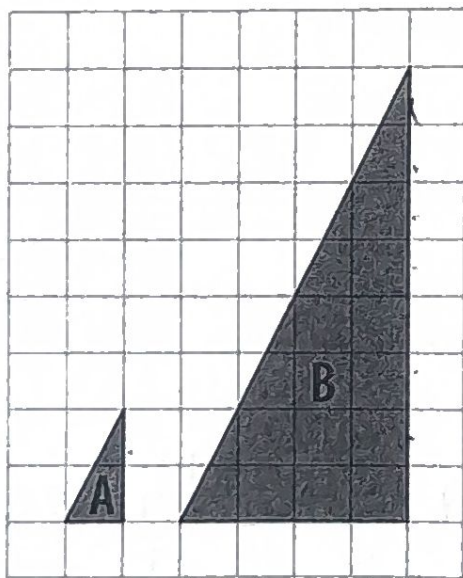
- b) How much does 1 cup of tea cost?

1 cake = £2

£3

[2 marks]

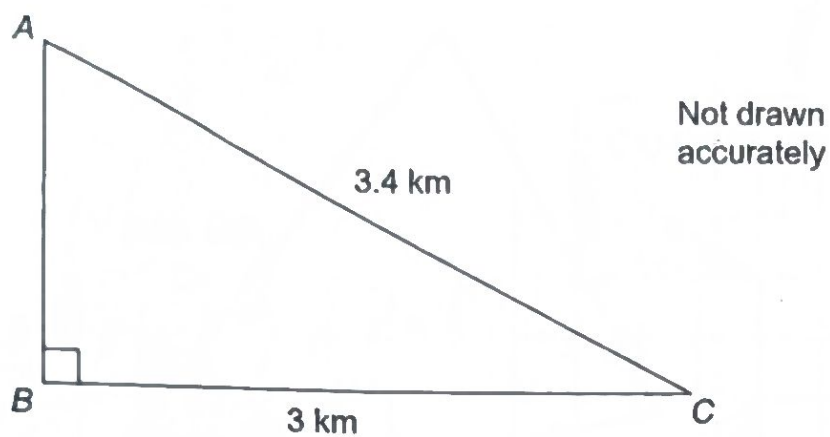
5. What is the scale factor of enlargement of shape A to shape B?



4

[1 mark]

6. The diagram shows three straight roads between villages A, B and C.



- a) Write down Pythagoras' theorem

$$a^2 + b^2 = c^2$$

[1 mark]

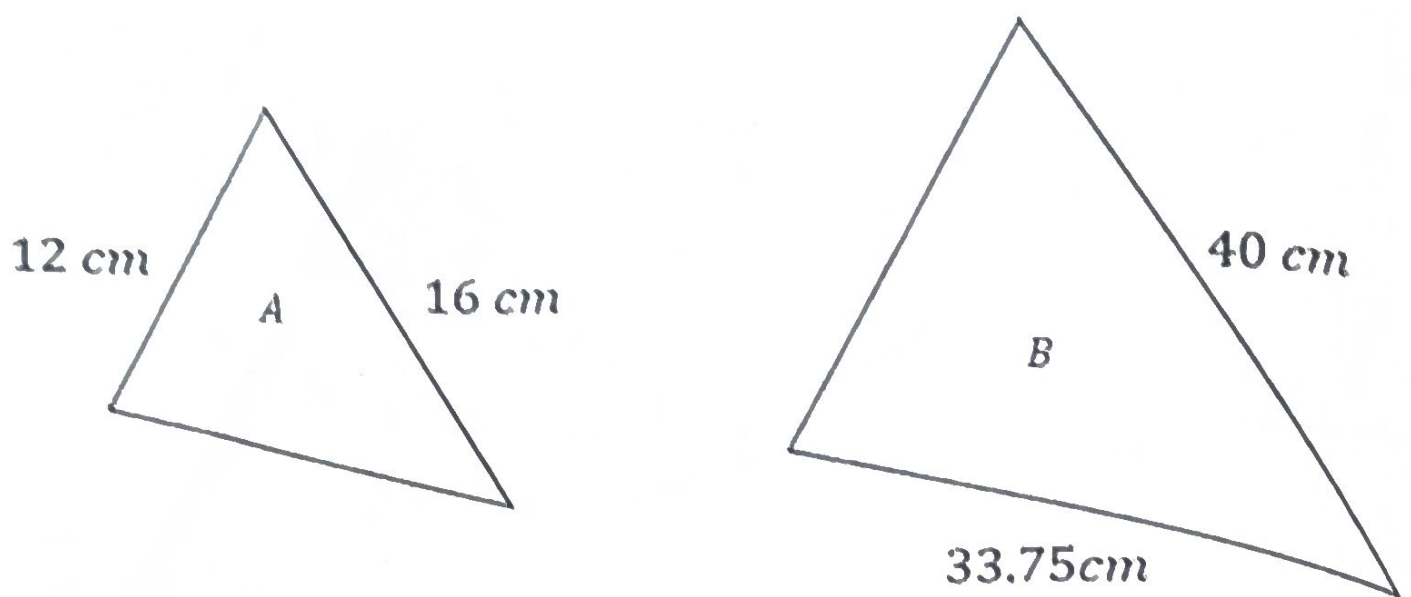
- b) Use Pythagoras to work out the distance AB.

$$\sqrt{3.4^2 - 3^2} = 1.6$$

1.6 km

[2 marks]

7. Here are two similar triangles.



Find the **perimeter** of triangle B

$$S.F = 2.5$$

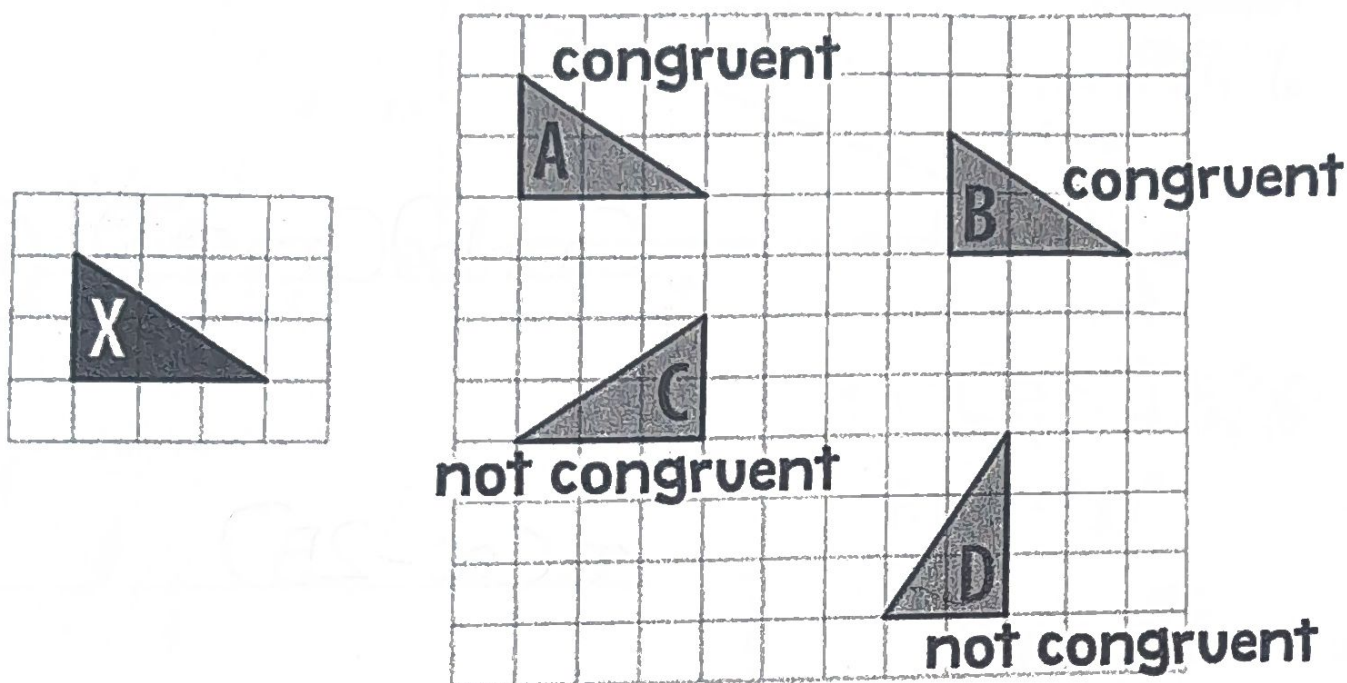
$$12 \times 2.5 = 30$$

$$30 + 33.75 + 40 = \underline{\underline{103.75}}$$

$$\underline{103.75 \text{ cm}}$$

[3 marks]

8. Charlie has sorted triangles A, B, C and D into those that are congruent to X and those that aren't.



- a) Explain what Charlie has done wrong.

~~Charlie state did not remember it was~~
 The ones charlie thought were not congruent [1 mark]
 were congruent, they were just turned

- b) Which triangles are congruent to X?

All of them

[2 marks]

$$x^2 + 2x + 5x + 10$$

$$x^2 + 7x + 10$$

SECTION 2:

1. Factorise fully:

a) $x^2 + 7x + 10$

$(x+5)(x+2)$ [2 marks]

b) $x^2 - 25$

$x(x-25)$ [1 mark]

c) $16m^3g^3 + 24m^2g^5$

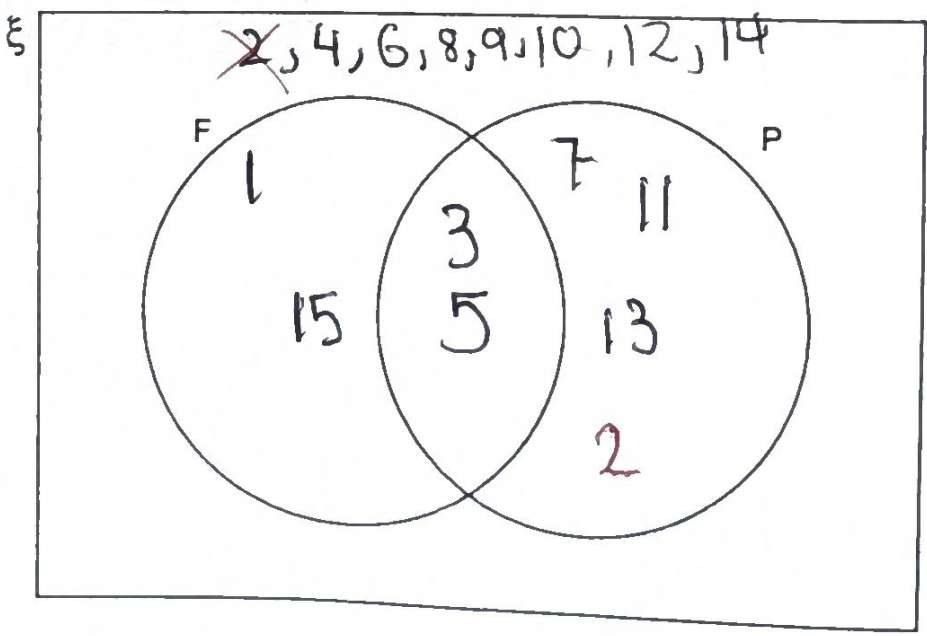
$4m^2g^3(4m+6g^2)$
 ~~$4m^2g^3(4m+6g^2)$~~
 ~~$16m^3g^3 + 24m^2g^5$~~ [2 marks]

2. Write the numbers from 1 to 15 in this Venn Diagram.

$\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$

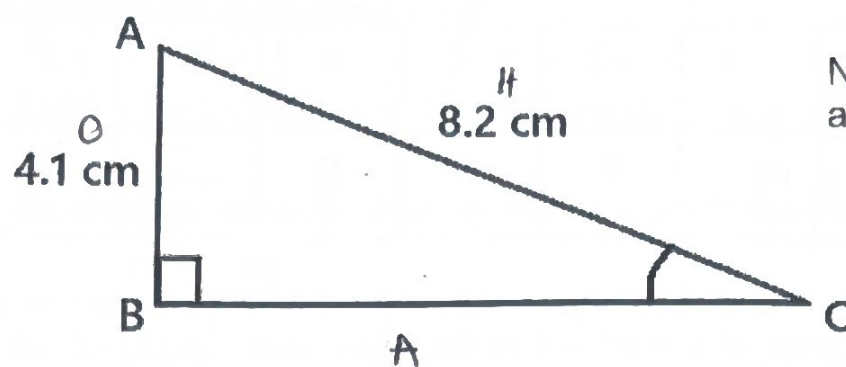
Set F = Factors of 15 {1, 3, 5, 15}

Set P = Prime numbers {2, 3, 5, 7, 11, 13}



3. a) Calculate the size of angle ACB.

SOHCAHTOA



Not drawn accurately

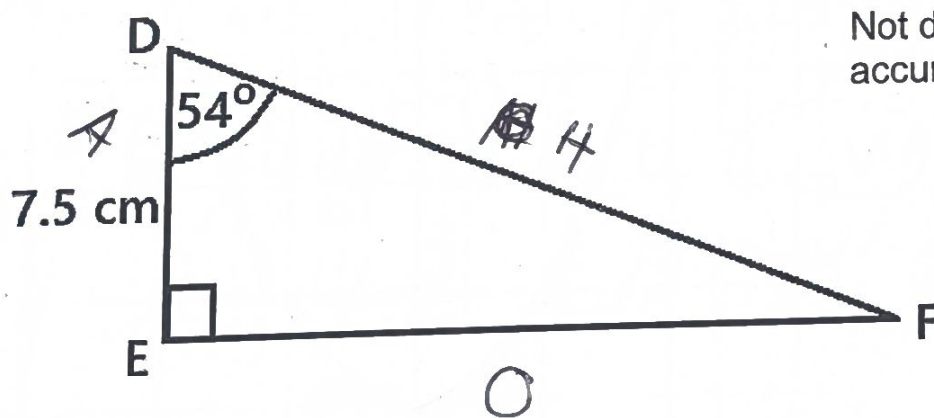
$$\sin^{-1}\left(\frac{4.1}{8.2}\right) = 30$$

$$\angle ACB = 30^\circ$$

30°

[3 marks]

b) Calculate the length DF correct to 3 significant figures.



Not drawn accurately

$$\cos(54) = \frac{7.5}{DF}$$

$$DF = \frac{7.5}{\cos(54)}$$

$$DF = \frac{7.5}{\cos(54)} = 12.7597 \dots$$

$$\underline{12.8 \text{ (3 s.f.)}}$$

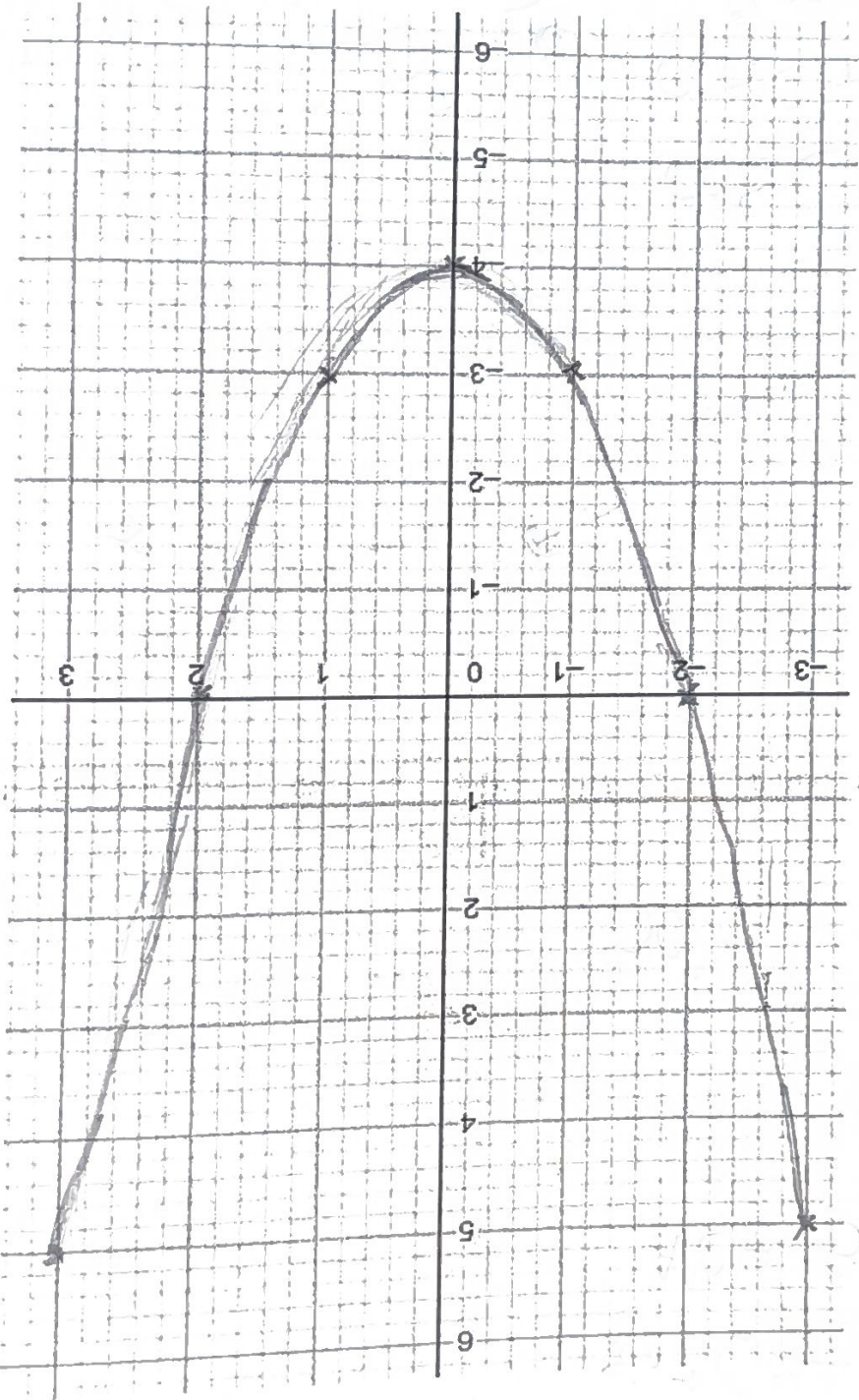
12.8

[3 marks]

4. a) Complete the table of values for $y = x^2 - 4$

x	y
-3	5
-2	0
-1	-3
0	-4
1	-3
2	0
3	5

b) Plot the graph of $y = x^2 - 4$ on the below axes



c) Use the graph to estimate the values of x when $y = 3$

~~$x = 2.6$~~
 ~~$x = -2.6$~~

$-2.6, 2.6$

[1 mark]

[4 marks]

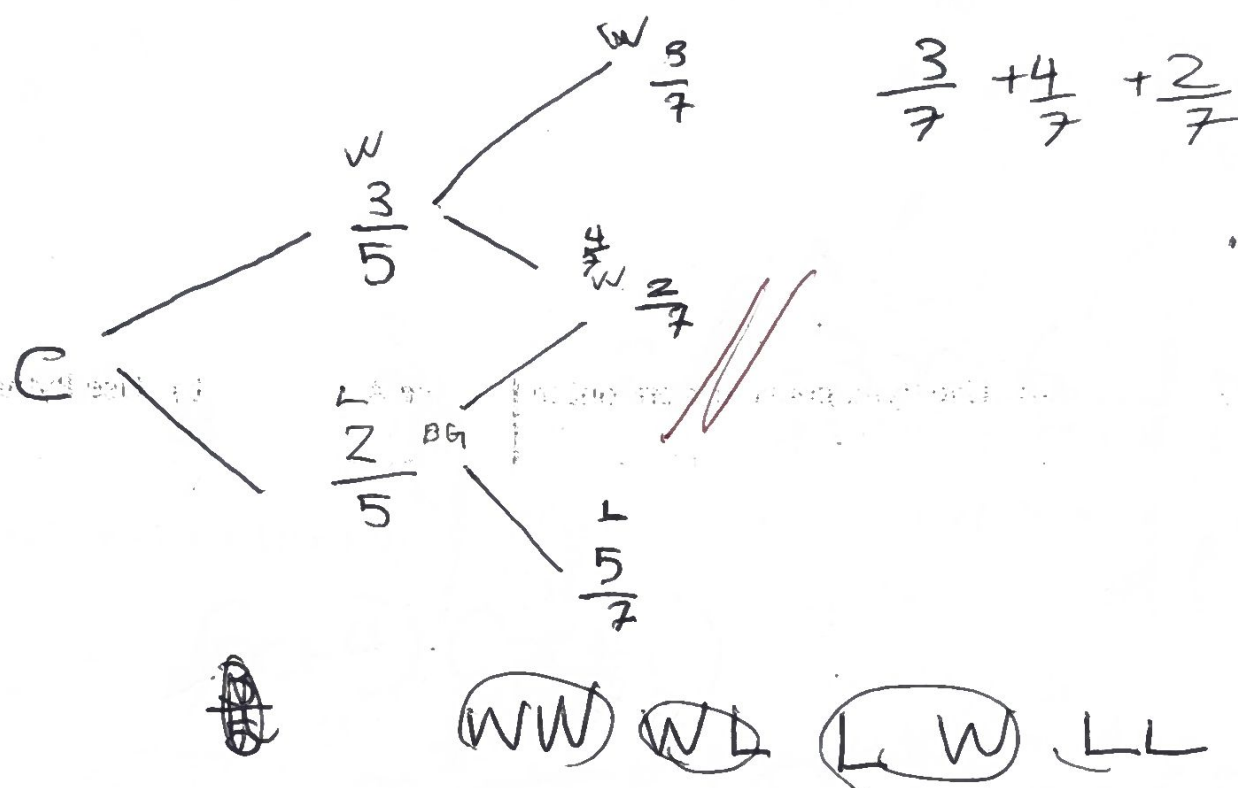
5. Each week Tom plays a game of chess and a game of backgammon.

The probability that he will win the chess game is $\frac{3}{5}$

When he wins the chess game, the probability that he will win the backgammon game is $\frac{3}{7}$

When he does not win the chess game, the probability that he will win the backgammon game is $\frac{2}{7}$

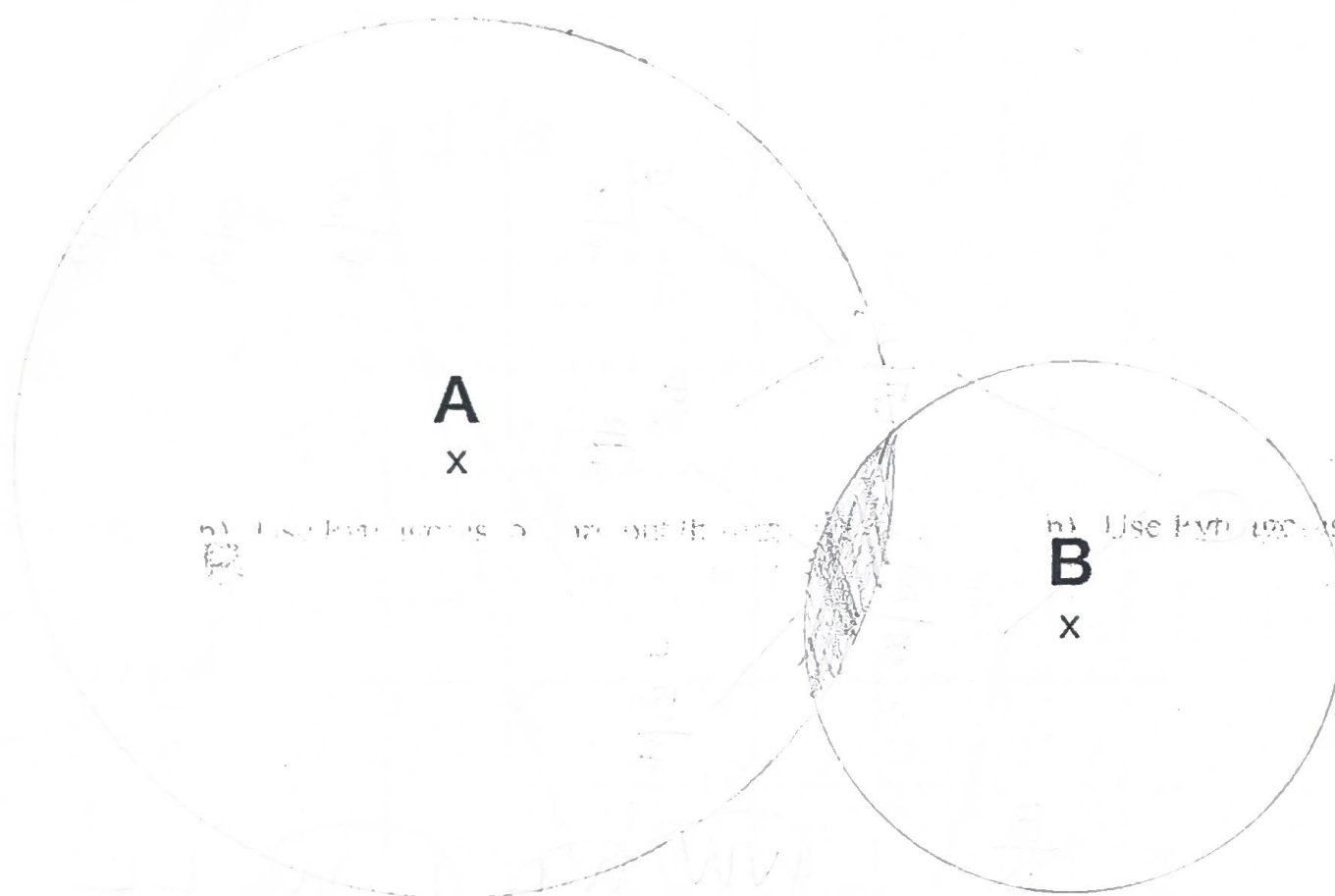
Draw a tree diagram to represent all the outcomes and use it to find the probability that he wins at least one of the games that he plays.



$$\frac{3}{4}$$

[4 marks]

6. Shade the locus of points that are less than 5 cm from A and less than 3 cm from B.



[3 marks]

SECTION 3:

1. Simplify $(2x + 3)^2 - (2x + 3)(x - 5)$

Give your answer in the form $ax^2 + bx + c$

$$4x + 9 - 2x^2 - 10x + 3x - 15$$

$$4x + 9 - 2x^2 - 7x - 15$$

$$2x^2 - 3x - 6$$

$$2x^2 - 3x - 6$$

[3 marks]

1

2. Expand $(x + 4)(x + 1)(x + 9)$

$$(x + 4)(x + 1)$$

$$x^2 + x + 4x + 4$$

$$x^2 + 5x + 4$$

$$x(x^2 + 5x + 4)$$

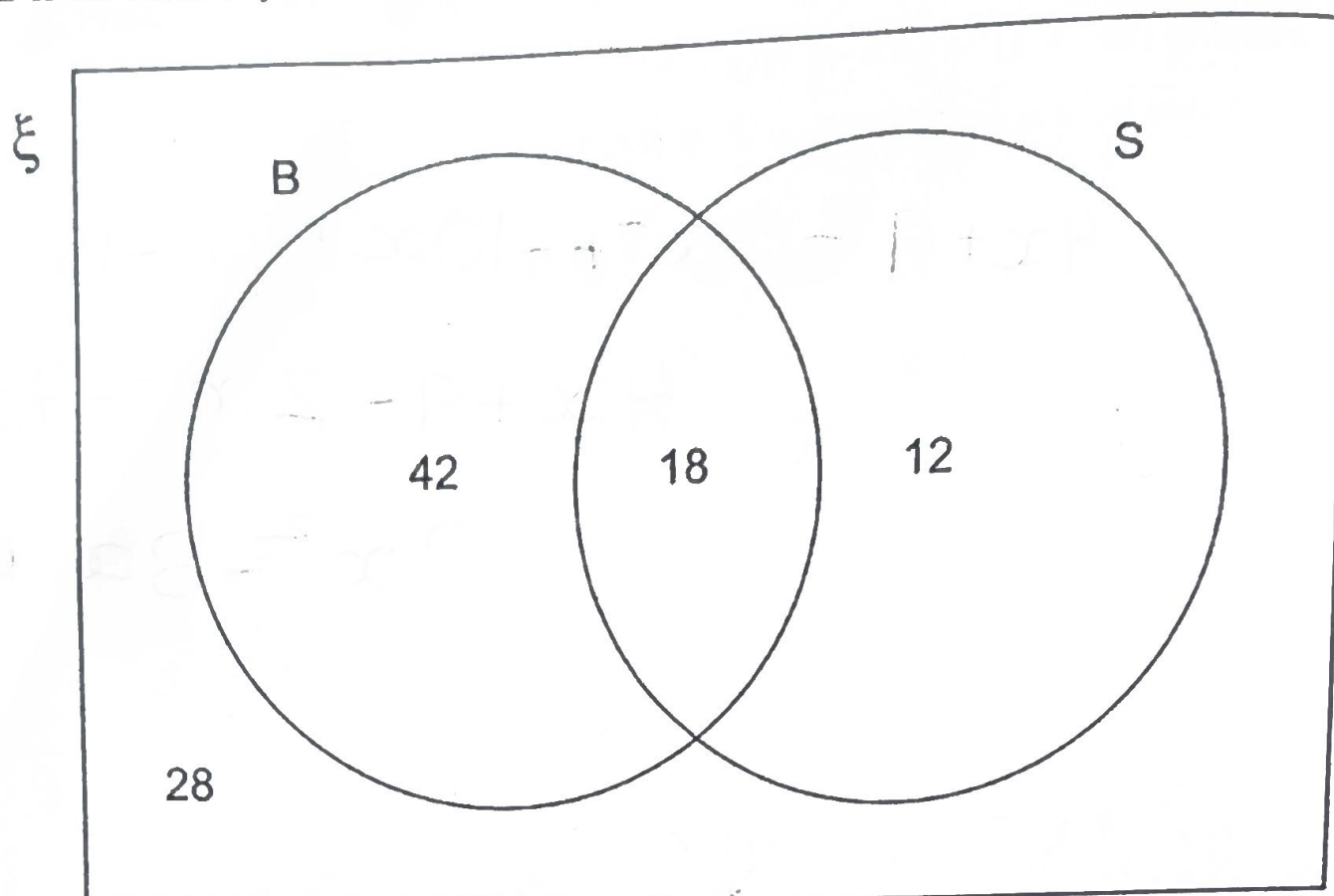
$$x^3 + 5x^2 + 4x$$

$$9x^3 + 14x^2 + 49x + 36$$

$$x^3 + 14x^2 + 49x + 36$$

[3 marks]

3. The Venn diagram shows information about 100 children.
B is the set of boys. S is the set of children who like skateboarding.



One child is chosen at random.

- a) Circle the value of $P(B')$

0.28

0.7

0.4

0.58

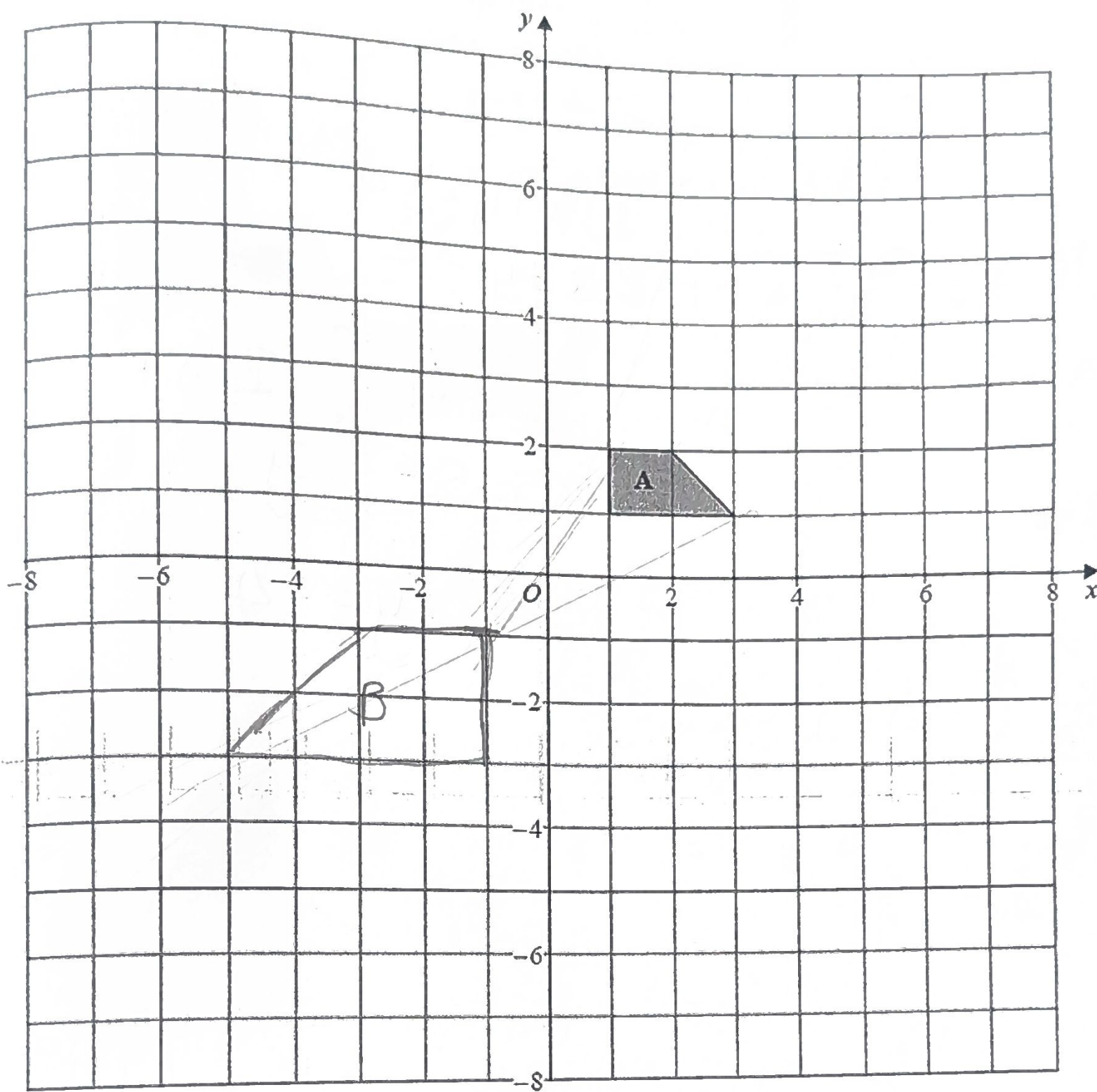
[1 mark]

- b) Use the values in the Venn diagram to show that $P(B \cap S) = P(B) \times P(S)$

$$P(18) = P(42) \times P(12)$$

[2 marks]

4. Look at the shape below:



a) Enlarge shape A by scale factor -2 , centre $(0, 0)$. Label your image B.

1 [2 marks]

b) Describe fully the single transformation that will map shape B onto shape A.

-2 SF Center $O O$

[2 marks]

5. Solve these simultaneous equations algebraically:

$$10x + y = 34$$

$$2x = 2 - y$$

$$\begin{array}{r}
 10x + y = 34 \\
 - \\
 2x + y = 2 \\
 \hline
 8x = 32 \\
 x = 4
 \end{array}$$

$$y = 8$$

$$x = 4$$

$$40 + y = 34$$

[4 marks]

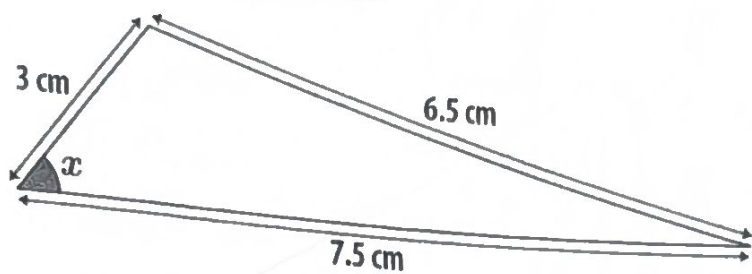
$$x = 4, y = -6$$

DO NOT WRITE IN THIS AREA

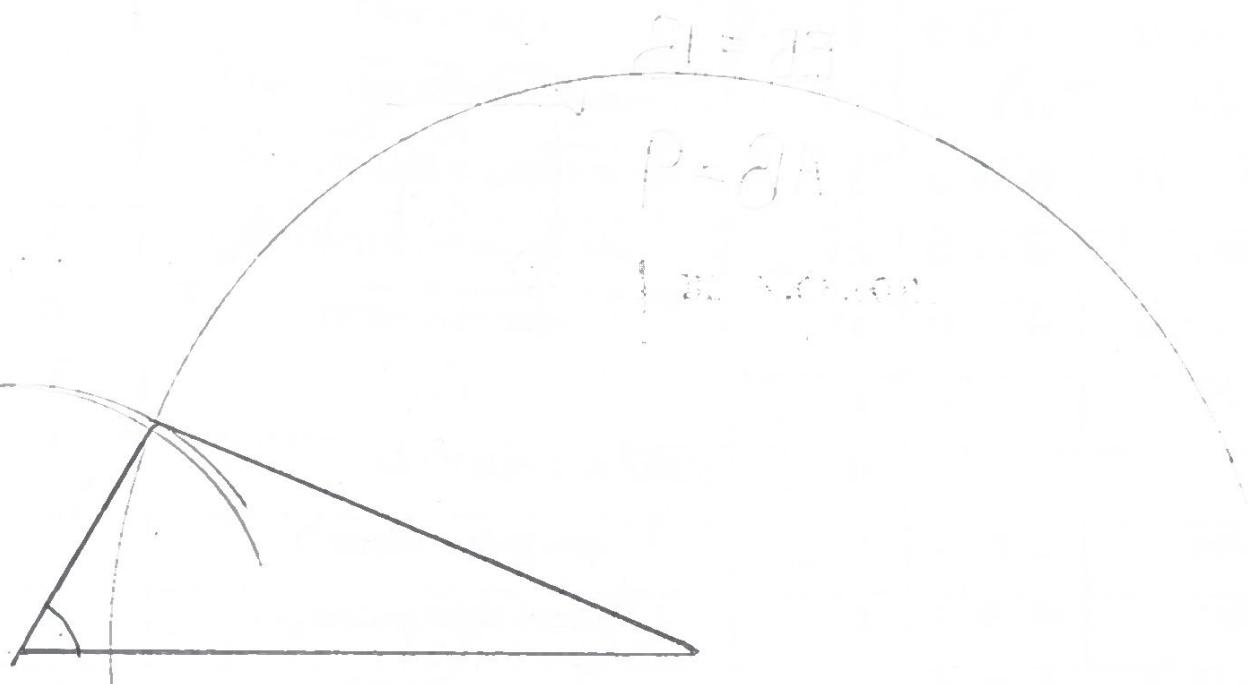
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

6. a) Using a ruler and a pair of compasses, make an accurate drawing of the triangle below. You must show all your construction lines.



Not drawn accurately



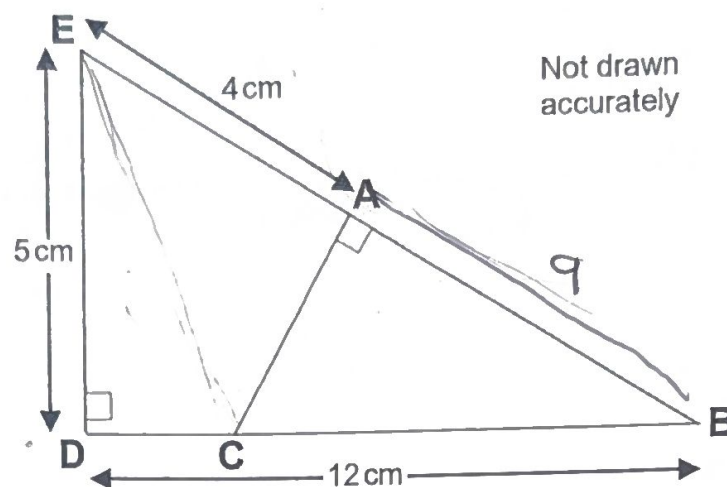
- b) Measure angle x in your diagram to the nearest degree.

[3 marks]

60°

[1 mark]

7. The diagram shows two right-angled triangles ABC and DEB.



Find the length of the line AC.

$$EB = 13$$

$$AB = 9$$

4 cm

2

[4 marks]

TOTAL FOR PAPER IS 75 MARKS

Question level analysis (complete after the assessment)

Qn no.	Topic	Mark	Rating	Sparx Codes
1	Algebra skills	— / 8	☹ ☹ ☹ ☹ ☹	M237, M792
2	Sample space	— / 2	☹ ☹ ☹ ☹ ☹	M718
3	Experimental probability	— / 2	☹ ☹ ☹ ☹ ☹	M332
4	Simple simultaneous with pictures	— / 3	☹ ☹ ☹ ☹ ☹	M852
5	Scale factors	— / 1	☹ ☹ ☹ ☹ ☹	M178
6	Pythagoras	— / 3	☹ ☹ ☹ ☹ ☹	M677, M480
7	Similar shapes	— / 3	☹ ☹ ☹ ☹ ☹	M377, M324
8	Congruence	— / 3	☹ ☹ ☹ ☹ ☹	M124
SECTION 1 SCORE: — / 25				
1	Factorising	— / 5	☹ ☹ ☹ ☹ ☹	M100, M908
2	Completing a Venn diagram	— / 2	☹ ☹ ☹ ☹ ☹	M829
3	Right-angled trigonometry	— / 6	☹ ☹ ☹ ☹ ☹	M212, M646
4	Plotting quadratic graphs	— / 5	☹ ☹ ☹ ☹ ☹	U989
5	Probability trees	— / 4	☹ ☹ ☹ ☹ ☹	M299, M572
6	Loci	— / 3	☹ ☹ ☹ ☹ ☹	M253
SECTION 2 SCORE: — / 25				
1	Complex expanding	— / 3	☹ ☹ ☹ ☹ ☹	M960
2	Expanding triple brackets	— / 3	☹ ☹ ☹ ☹ ☹	U606
3	Venn diagrams	— / 3	☹ ☹ ☹ ☹ ☹	M419
4	Enlargement	— / 4	☹ ☹ ☹ ☹ ☹	M178
5	Simultaneous equations	— / 4	☹ ☹ ☹ ☹ ☹	M852, M658
6	Constructing triangles	— / 4	☹ ☹ ☹ ☹ ☹	M565
7	Problem solving with triangles	— / 4	☹ ☹ ☹ ☹ ☹	M324, M480
SECTION 3 SCORE: — / 25				